



Advanced Digital Image Processing Lecture #9

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Contents

- ❖ **Pattern Recognition**
- ❖ Template Matching
- ❖ PCA Method
- ❖ Subspace Method
- ❖ Knowledge-Based Approach
- ❖ Neural Network
- ❖ Hidden Markov Model (HMM)



Introduction: Pattern Recognition

- ❖ A pattern is an arrangement of descriptors (denoted as feature) or individual image regions (objects).
- ❖ A pattern class is a family of patterns that share some common properties
- ❖ Pattern recognition refers to the classification of objects or patterns



Requirements for Pattern Recognition Systems

- ❖ The design of pattern recognition systems requires that a set of training patterns, which are patterns with extrinsic pattern class labels be available
- ❖ Central theme of recognition is the concept of “learning” from sample pattern
- ❖ Formation of decision rules for pattern recognition or classification.



Pattern Learning Methods

- ❖ Supervised Learning: labeled training samples
- ❖ Unsupervised Learning: unlabeled training samples
- ❖ Semi-supervised Learning: labeled with few samples and then adapt more unlabeled samples



Example of Pattern Recognition Systems

- ❖ Commercial pattern recognition systems are available for,
 - ☐ Optical character recognition,
 - ☐ Face recognition,
 - ☐ Speech recognition,
 - ☐ Speaker recognition,
 - ☐ Finger print recognition, etc.



Basic Approaches for Pattern Recognition

- ❖ Statistical Approach
- ❖ Structural or Geometrical
- ❖ Knowledge-Based
- ❖ Neural Network
- ❖ Hybrid Technology

Pattern Recognition: Statistical

- ❖ Statistical: The features are assumed to have a probability density function condition on the pattern class.
- ❖ Thus a pattern vector x belonging to class w_j is viewed as an observation drawn randomly from the class condition density, $p(x|w_j)$, where, $j=1, \dots, k$

Pattern Recognition: Structural

- ❖ Not only quantitative measures about each feature but also the spatial relationships between the features determine class membership. Example,
- ❖ Finger print recognition: is based on the interrelationships of print features called minutiae. Together with their relative sizes and locations these features are primitive components that describe fingerprint ridge properties, such as abrupt endings, branching, merging and disconnected segment

Template Matching

- ❖ Correlation coefficient

$$\alpha_t = M_t / P_t (0 < \alpha_t \leq 1)$$

- ❖ Manhattan distance

$$\delta_t = \left\{ \sum_1^{x \times y} |I - G_t| \right\}$$

- ❖ Where M_t is the total number of matched pixels and P_t is the total number of pixels. $I(x,y)$ input image and $G_t(x,y)$ is t^{th} template image.



Multiple Feature Based Template Matching

- ❖ Use multiple features:
 - ❖ Correlation coefficient
 - ❖ Minimum distance
- ❖ If two methods classify the image into the same class then the pose is recognized; otherwise ignored.



Template Matching

- ❖ Image and Template are the same sizes (same resolution)
- ❖ Object size (in an image) greater/smaller according to camera and object distance, in that case,
 - ❖ Multi-resolution templates or template pyramid are used
 - ❖ Or, Original image is resized multiple times.

Example of Templates

Sample Templates
(60x60)

Test image
(60x60)

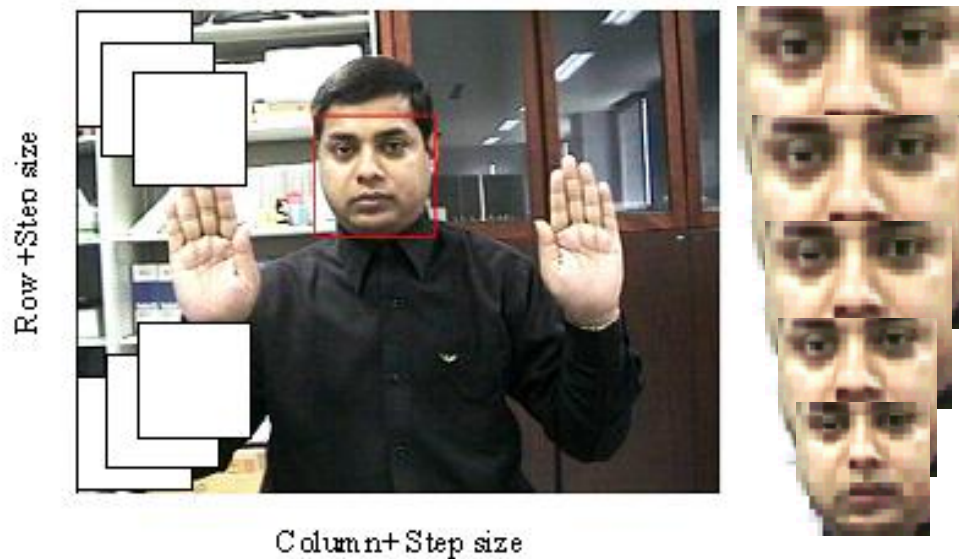




Face Detection Using Multi-resolution Templates

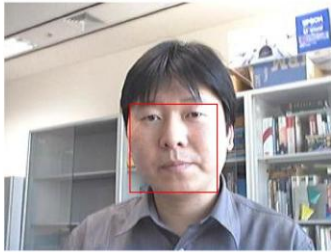
- Step 1: Prepare Template images with different resolutions
- Step 2: For each frame template image sliding starts from the (0,0) position of the image and progresses it by a given step size from left to right and top to bottom.
- Step 3: Measure Minimum distance or Correlation Coefficient
- Step 4: This process is done until template reaches the end of the input image
- Step 5: Based on specific threshold detect face area and draw a boundary.

Face Detection Using Multi-resolution Templates

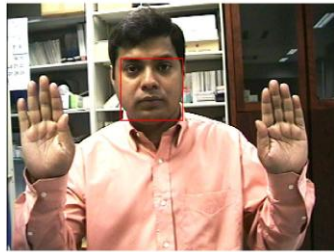


- ❖ This method uses the template images of (50X50), (60X60), (70X70), (80X80), (90X90), and (100x100) dimensions for face detection.

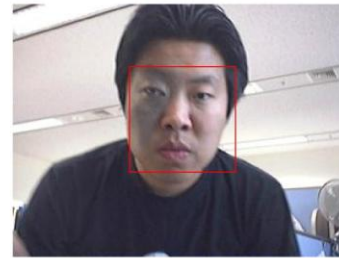
Sample Outputs of Face Detection



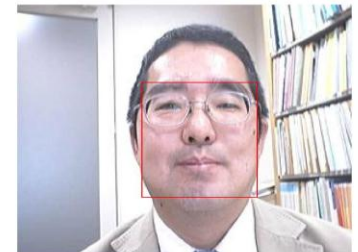
(a)



(b)



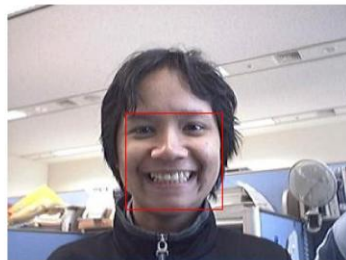
(c)



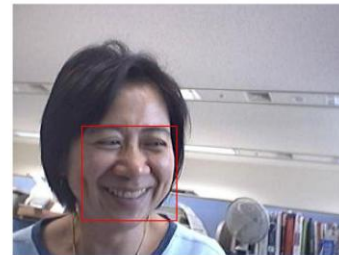
(d)



(e)



(f)



(g)



(h)



Assignment for Next Week (Student-Group1)

- ❖ Explain following terms using examples:
 - ❖ Mean, median
 - ❖ Variance
 - ❖ Covariance
 - ❖ Eigenvectors
 - ❖ Eigenvalues
 - ❖ PCA (principal component)



Questions? Or Suggestions?

Thanks to all